Fernando Lucatelli Nunes

Researcher in Mathematics and Computer Science

Appointed Assistant Professor, University of Coimbra

Research profile: Programming Languages, Logic, and Semantics, with deep foundations in categorical structures.

arXiv preprints: `https://arxiv.org/a/lucatellinunes_f_1.html`

Google Scholar

Scopus ID 57188838707

Mathematics Genealogy

Webpage: https://flnlucatelli.github.io/ **E-mail:** fernandolucatellinunes@gmail.com

PhD IN MATHEMATICS (AWARDED 2018)

10/2012 – 01/2018 **University of Coimbra**, PhD in Mathematics

- Thesis: *Pseudomonads and Descent* (awarded *summa cum laude*).
- Supervisor: Prof. Maria Manuel Clementino.
- Examination Committee: Prof. Martin Hyland, Prof. Marco Mackaay, Prof. Lurdes Sousa, and Prof. Peter Beier Gothen.
- Main contributions: Biadjoint triangle theorems; freely generated categorical structures; theories of presentation and coherence for low- and higher-dimensional categorical structures; Grothendieck descent theory; categorical Galois theories; low- and higher-dimensional weak limits.

PhD IN COMPUTER SCIENCE (Expected April 2026)

11/2020 – 11/2025 **Utrecht University**, PhD in Computer Science

- Thesis: *Freely Generated Categorical Structures and Automatic Differentiation.*
- Supervisors: Prof. Gabriele Keller and Dr. Matthijs Vákár.
- Examination Committee: Prof. Ieke Moerdijk, Prof. Bart Jacobs, Prof. Sam Staton, Prof. Robin Cockett, and Prof. Patricia Johann.
- Main contributions: Denotational semantics of programming languages; automatic differentiation; principled program transformations; recursion and iteration; categorical semantics; logical relations; type systems and dependent types; Grothendieck constructions; extensive categories.

Academic Positions

ACADEMIC APPOINTMENTS

2025 – present	**Research Fellow**, School of Computer Science, University of Birmingham, UK.
2025	**Postdoctoral Fellow**, Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2023	**Postdoctoral Fellow**, York University, Canada.
2023	**Fields Research Fellow**, Fields Institute for Research in Mathematical Sciences, Canada.
2022	**Oberwolfach Leibniz Fellow**, Mathematisches Forschungsinstitut Oberwolfach, Germany.
2018 – 2021	**Research Fellow**, University of Coimbra, Portugal.
2018 – 2019	**Invited Assistant Professor**, University of Coimbra, Portugal.
2018	**Postdoctoral Fellow**, Centre for Mathematics, University of Coimbra, Portugal.

VISITING ACADEMIC APPOINTMENTS

2018, 2023, 2024, 2025	**Visiting Researcher**, UCLouvain, Belgium.
2025	**Visiting Researcher**, IMPA, Rio de Janeiro, Brazil.
2019, 2025	**Visiting Researcher**, University of Cape Town, South Africa.
2019, 2025	**Visiting Researcher**, Stellenbosch University, South Africa.

RESEARCH GROUPS AND AFFILIATIONS

2025 – present	**Theory of Computation**, School of Computer Science, University of Birmingham, UK.
2020 – 2026	**Software Technology Group**, Utrecht University, The Netherlands.
2020 – 2024	**External Researcher**, Centre for Mathematics (CMUC), University of Coimbra, Portugal.
2013 – present	**Coimbra Category Theory Group**, University of Coimbra, Portugal.

DOCTORAL APPOINTMENTS

2020 – 2025	**Promovendus**, Utrecht University, The Netherlands.
2012 – 2018	**Doctoral Researcher**, University of Coimbra, Portugal.

EARLY ACADEMIC EXPERIENCE

2006 – 2008	**Secondary Mathematics Teacher**, Curso Dínatos, Brasília, Brazil.

Research

RESEARCH INTERESTS

Expertise and research interests: Mathematics and Computer Science, centred on the semantic and structural foundations of computation: (1) programming language semantics and formal methods, including rewriting systems, program analysis, and principled program transformations; (2) low- and higher-dimensional universal algebra and generalised (multi)categorical structures; (3) categorical methods in logic, topology, and algebra.

SELECTED PUBLICATIONS

Lucatelli Nunes, F., Vákár, M.: **Monoidal closure of Grothendieck constructions via Σ-tractable monoidal structures and Dialectica formulas.** Theory Appl. Categ. 44, 1153–1217. 2025. (*65 pages*).
Lucatelli Nunes, F., Vákár, M.: **Automatic differentiation for ML-family languages: Correctness via logical relations.** Mathematical Structures in Computer Science. 34, No. 8, 747–806 (60 pages).
Lucatelli Nunes, F., Vákár, M.: **CHAD for Expressive Total Languages.** Math. Struct. Comput. Sci. 33, No. 4–5, 311–426. 2023. (*116 pages*).
Lucatelli Nunes, F., Prezado, R., Sousa, L.: **Cauchy Completeness, Lax Epimorphisms, and Effective Descent for Split Fibrations.** Bull. Belg. Math. Soc. - Simon Stevin 30, No. 1, 130–139. 2023. (*10 pages*).
Lucatelli Nunes, F.: **Semantic factorization and descent.** Appl. Categ. Struct. 30, No. 6, 1393-1433. 2022. (*41 pages*).
Lucatelli Nunes, F., Sousa, L.: **On lax epimorphisms and the associated factorization.** J. Pure Appl. Algebra 226, No. 12, 27 p. 2022. (*27 pages*).
Lucatelli Nunes, F.: **Pseudo-Kan Extensions and Descent Theory.** Theory Appl. Categ. 33, 390–444. 2018. (*35 pages*).
Lucatelli Nunes, F.: **On biadjoint triangles.** Theory Appl. Categ. 31, 217–256. 2016. (*40 pages*).

PREPRINTS

Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Unraveling the iterative CHAD.** arXiv:2505.15002. 2025. (*57 pages*).
Lucatelli Nunes, F., Prezado, R.: **Functors Preserving Effective Descent Morphisms.** arXiv:2410.22876. 2024. (*17 pages*).
Lucatelli Nunes, F., Vákár, M.: **Free doubly-infinitary distributive categories are cartesian closed.** arXiv:2403.10447. 2024. (*16 pages*).
Lucatelli Nunes, F.: **Freely generated n-categories, coinserters and presentations of low dimensional categories.** arXiv:1704.04474. 2017. (*56 pages*).

OTHER JOURNAL PUBLICATIONS

Prezado, R., Lucatelli Nunes, F.: **Generalized multicategories: change-of-base, embedding, and descent.** Appl. Categ. Struct. Volume 32, 35. 2024. (*89 pages*).
Lucatelli Nunes, F., Prezado, R., Vákár, M.: **Free extensivity via distributivity.** Portugaliae Mathematica, 82, no. 1/2, pp. 177–204. 2025. (*28 pages*).
Clementino, M.M., Lucatelli Nunes, F., Prezado, R.: **Lax comma categories: cartesian closedness, extensivity, topologicity, and descent.** Theory Appl. Categ. 41, 516–530. 2024. (*15 pages*).
Clementino, M.M., Lucatelli Nunes, F.: **Lax comma 2-categories and admissible 2-functors.** Theory Appl. Categ. 40, 180–226. 2024. (*47 pages*).
Clementino, M.M., Lucatelli Nunes, F.: **Lax comma categories of ordered sets.** Quaest. Math. 46, Part Suppl. 1, 145–159. 2023. (*14 pages*).
Prezado, R., Lucatelli Nunes, F.: **Descent for internal multicategory functors.** Appl Categ. Struct 31, 11. 2023. (*18 pages*).
Lucatelli Nunes, F.: **Descent data and absolute Kan extensions.** Theory Appl. Categ. 37, 530–561. 2021. (*32 pages*).
Lucatelli Nunes, F.: **Pseudoalgebras and non-canonical isomorphisms.** Appl. Categ. Struct. 27, No. 1, 55–63. 2019. (*9 pages*).
Lucatelli Nunes, F.: **On lifting of biadjoints and lax algebras.** Categ. Gen. Algebr. Struct. Appl. 9, No. 1, 29–58. 2018. (*30 pages*).
Lucatelli Nunes, F.: **Espaços não reversíveis.** (Non-reversible spaces). Revista Matemática Universitária, v. 48, n. 49, p. 22-26. 2012. (*5 pages*).

WORKSHOP AND OTHER PUBLICATIONS

Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Homomorphic Reverse Differentiation of Iteration.** LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024). 2024. (*2 pages*).
Lucatelli Nunes, F., Vákár, M.: **Logical Relations for Partial Features and Automatic Differentiation Correctness.** arXiv:2210.08530. 2022. and Oberwolfach Preprints, OWP-2023-09. 2023. (*21 pages*).
Lucatelli Nunes, F.: **Pseudomonads and Descent.** PhD Thesis, University of Coimbra. 2017. (*209 pages*).

UNDERGRADUATE AND GRADUATE MONOGRAPHS

Lucatelli Nunes, F.: **Basic Concepts of Topology for Topological Dynamics (in Portuguese).** 2009. (*116 pages*).
Lucatelli Nunes, F.: **Basic Topological Dynamics and Basic Applications to Number Theory (in Portuguese).** 2009. (*57 pages*).

Grants and Fellowships

RESEARCH FELLOWSHIPS (PI)

2022 — **Oberwolfach Leibniz Fellowship**, Grothendieck Descent Theory, Galois Theory, and Presentations of Categorical Structures

- PI: **Fernando Lucatelli Nunes**.
- Participants: Maria Manuel Clementino, Rui Prezado, Lurdes Sousa, Matthijs Vákár.

2023 — **Fields Research Fellowship**, Grothendieck Descent Theory, Fibrations, and Factorizations

- PI: **Fernando Lucatelli Nunes**.
- Participants: Walter Tholen.

SELECTED PARTICIPATION IN RESEARCH PROJECTS AND GRANTS

2025 – present — **ERC Starting Grant FoRECAST**, Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy

- PI/Grant holder: Matthijs Vákár.
- My role: Member of the formal methods team, contributing to the design of principled algorithms grounded in categorical structures, and involved in the (co-)supervision of PhD students.

2025 – present — **DFG Project HOMBRe**, Higher-Order Monad-based Programming and Reasoning

- PI/Grant holder: Sergey Goncharov.
- My role: Postdoctoral Fellow, contributing to the core research themes of the project.

2021 – 2024 — **NWO Veni Grant (VI.Veni.202.124)**, Correct Expressive Differential Programming: Laying the Mathematical Foundations for Tomorrow's Machine Learning

- PI/Grant holder: Matthijs Vákár.
- My role: Led the work on Iterative CHAD, contributing to the formal methods component of the project. Additionally, established the correctness of dual-numbers-based (forward- and reverse-mode) automatic differentiation for languages with recursive types.

2020 – 2022 — **Marie Skłodowska-Curie Grant**, Principled AD Program Transformations

- PI/Grant holder: Matthijs Vákár.
- My role: Developed the correct extension of reverse-mode automatic differentiation (CHAD) program transformations to expressive total languages.

2011 – 2011 — **Project: Nonlinear Analysis and Applications (CNPq)**, Group Project (University of Brasília), funded by CNPq

- Project title: Nonlinear Analysis and Applications.
- My role: Undergraduate student; led the subproject "Topological Dynamics and Applications in Number Theory" and authored two monographs on the topic.

Selected Graduate Supervision and Mentorship

PhD STUDENTS (THESIS SUPERVISION)

2026 – present

Utrecht University, Department of Information and Computing Sciences

- Student: **Diogo Simm**.
- Research area: Programming language semantics and category theory.
- Co-supervision with Dr. Matthijs Vákár and Prof. Gabriele Keller.
- Funded under the ERC Starting Grant FoRECAST.

2025 – present

University of Coimbra, Joint University of Coimbra–University of Porto PhD Programme in Mathematics

- Student: **Néot Choserot**.
- Research area: Generalised categorical structures.
- Funded by the Centre for Mathematics, University of Coimbra.

2020 – 2024

University of Coimbra, Joint University of Coimbra–University of Porto PhD Programme in Mathematics

- Student: **Rui Rodrigues de Abreu Fernandes Prezado**.
- Thesis: *Some Aspects of Descent Theory and Applications*.
- Research area: Generalised multicategories, and Grothendieck descent theory.
- Co-supervision with Prof. Maria Manuel Clementino.
- Funded by FCT, Portugal.
- PhD awarded *summa cum laude* (January 2024).

MASTER'S STUDENTS (THESIS SUPERVISION)

2025

IMPA, Rio de Janeiro, Brazil, Master's Programme in Pure Mathematics

- Student: **Diogo Simm**.
- Co-supervised with Prof. Henrique Bursztyn and Dr. Matthijs Vákár.
- Research area: Programming language semantics, automatic differentiation, and category theory.
- Degree awarded with distinction (grade A).

RESEARCH-BASED MASTER'S PROJECT SUPERVISION

2025

Utrecht University, The Netherlands, Scientific Methods in Computing Science (Programming Languages)

- Supervision of multiple small-group research projects at Master's level.
- Topics included symbolic and automatic differentiation, program transformations, and categorical semantics.
- Representative projects:
  - *Calculating the Symbolic Derivative using Automatic Differentiation.*
  - *Lambda-Set Specialisation for Haskell.*
  - *Dual-Numbers Automatic Differentiation with Inductive Types.*

Selected Undergraduate Supervision

UNDERGRADUATE THESIS AND CAPSTONE SUPERVISION

2025 – present

Utrecht University, BSc Thesis in Artificial Intelligence

- Student: **Sander Santema**.
- Thesis topic: A local defunctionalisation transformation for the simply typed lambda calculus.
- Areas: Categorical semantics, program transformation, defunctionalisation, functional programming, Haskell, and generalised algebraic data types.

2023

Utrecht University, BSc Capstone Project in Computer Science

- Students: **Cérine Touber**, **Oscar Mensink**, **Kasper Peeters**, **Nigel Strachan**, **Joost Laan**, **Jippe Hoogeveen**, **Gust Verhees**, **Nasr Antar**, and **Artur Sałek**.
- Project: Design and implementation of a software system for identifying optimal collaboration matches within organisations, based on interaction data and weighted optimisation criteria.
- Client: Dr. Ioanna Lykourentzou
- Areas: Optimisation algorithms, software engineering, and web-based systems.

UNDERGRADUATE SOFTWARE PROJECTS

2021–2022

Utrecht University, Game Technology

- Supervision of nine undergraduate students organised into three project teams.
- Project: Design and development of a game implemented in C#.

Teaching and Public Engagement

SELECTED TEACHING EXPERIENCE

2025 – 2026 **University of Coimbra, Portugal**, BSc program in Computer Engineering and in Artificial Intelligence and Data Science

- **Main Lecturer**.
- Course: *Mathematical Analysis II.*

2025 – 2026 **University of Coimbra, Portugal**, Joint PhD Programme in Mathematics (Coimbra–Porto)

- **Main Lecturer**.
- Course: *Categories in Algebra and Topology.*
- Topics include enriched categories, monads, and generalised multicategories.

2020 – 2025 **Utrecht University, The Netherlands**, Department of Information and Computing Sciences

- **Lecturer and Graduate Teaching Assistant (PhD appointment).**
- Functional Programming — tutorials and assessment.
- Logic for Computer Science — tutorials and project supervision.
- Graphics for Computer Science (including linear algebra components).
- Supervision of BSc Computer Science capstone software projects.
- Coaching and supervision of undergraduate Game Technology projects.
- **Recipient of the Utrecht University Distinguished Teaching Award (2023).**

2023 – 2024 **Utrecht University, The Netherlands**, Department of Information and Computing Sciences

- **Main Lecturer**, BSc Computer Science.
- Course: *Logic for Computer Science.*
- Topics included inference systems, natural deduction, semantics, and Hoare logic.

2020 **University of Porto, Portugal**, Joint PhD Programme in Mathematics (Coimbra–Porto)

- **Main Lecturer**.
- Course: *Categories in Algebra and Topology.*
- Topics included enriched categories, monads, and generalised multicategories.

2018 – 2019 **University of Coimbra, Portugal**, Department of Mathematics

- **Invited Assistant Professor**.
- Course: *Topology and Linear Analysis* (co-taught with Prof. Maria Manuel Clementino).

2017 – 2020 **University of Coimbra, Portugal**, Department of Mathematics

- **Instructor, Delfos Project**.
- Advanced mathematics teaching for secondary-school students preparing for national and international mathematics competitions.

2008 – 2012 **University of Brasília, Brazil**, Department of Mathematics

- **Undergraduate Teaching Assistant**.
- Courses included differential and integral calculus, real analysis, linear algebra, and algebra I.

2006 – 2008 **Curso Dínatos, Brasília, Brazil**, Secondary Education

- **Secondary Mathematics Teacher**.
- Mathematics instruction for students preparing for national university entrance examinations.

2005 – 2006 **EPCAR, Brazilian Air Force, Brazil**, Secondary Education
- **Student Teaching Assistant**.
- Mathematics instruction for pre-cadet students.

OUTREACH AND PUBLIC ENGAGEMENT

2017 – 2020 **Delfos Project**, Department of Mathematics, University of Coimbra, Portugal
- Contributor to the Delfos Project, a mathematics education programme for secondary-school students preparing for international mathematics competitions.
- Served as Portuguese representative to the International Mathematical Olympiad (IMO) in 2020.

2008 – 2012 **Scientific Initiation (CNPq)**, Brazil
- Undergraduate member of the research group *Teoria de Lie e Dinâmica*.
- Contributed to the outreach blog `https://liedinamica.wordpress.com/`.
- Authored three educational monographs in Portuguese on Lie groups, topology for dynamical systems, and applications of topological dynamics to number theory.
- Published an expository article in *Revista Matemática Universitária* (Portuguese).

2011 – 2012 **Brazilian Society for Scientific Progress (SBPC)**, Brazil
- Presented multiple popular-science posters in mathematics at national SBPC conferences.
- Topics included game theory, homotopy theory, category theory, topology, and dynamical systems.
- All contributions were published in the conference annals (in Portuguese).

Evaluation Committees

DOCTORAL EXAMINATION COMMITTEES

2024

PhD Defense Committee, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal

- Candidate: **Rui Rodrigues de Abreu Fernandes Prezado**.
- Thesis: *Some Aspects of Descent Theory and Applications*.
- Role: Supervisor and examiner.

2023

PhD Defense Committee, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal

- Candidate: **Carlos Miguel Alves Fitas**.
- Thesis: *On Lax Idempotent Monads in Topology*.
- Role: Main examiner and opponent.

DOCTORAL PROJECT EVALUATION

Member of doctoral project evaluation committees within the Joint University of Coimbra–University of Porto PhD Programme in Mathematics (2019), assessing research proposals in category theory and topology, including projects on lax idempotent monads and Grothendieck descent theory. Participated as examiner and supervisor in the formal evaluation of doctoral research plans.

OTHER ACADEMIC EVALUATIONS

2025

Bachelor's Thesis Proposal Evaluation, BSc in Artificial Intelligence, Utrecht University, The Netherlands

- Role: Second reader.
- Topic: *Generative AI as a Study Tool in Higher Education*.

Academic Service and Governance

RECRUITMENT AND GOVERNANCE

2025 — **PhD Recruitment Committee Member**, ERC Starting Grant FoRECAST

- Member of the hiring committee for three PhD positions under the ERC Starting Grant project *Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy.*

2023 – 2025 — **Member, Informatics Research Advisory Committee (IRAC)**, Utrecht University, The Netherlands.

2020 — **Programme Committee Member**, Joint University of Coimbra–University of Porto PhD Programme in Mathematics, Portugal.

EDITORIAL AND REFEREEING ACTIVITY

Referee for leading international journals and conferences in mathematics and computer science, including but not limited to: *ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*, *Advances in Mathematics*, *Applied Categorical Structures*, *Homology, Homotopy and Applications*, *Journal of Pure and Applied Algebra*, *Mathematical Structures in Computer Science*, *Order*, *Quaestiones Mathematicae*, *Theory and Applications of Categories*, and *Topology and its Applications*.

ORGANISATION OF SCIENTIFIC EVENTS AND SEMINARS

2025 – present — **Organiser, Online Algebra, Logic and Topology Seminar**, University of Coimbra, Portugal.

2019 – 2023 — **Organiser, Algebra, Logic and Topology Seminar**, University of Coimbra, Portugal.

2022 — **Organising Committee Member, XIII Portuguese Category Seminar**, University of Coimbra, Portugal.

2019 — **Organising Committee Member, XII Portuguese Category Seminar**, University of Coimbra, Portugal.

2018 — **Organiser, Workshop on Algebra, Logic and Topology**, University of Coimbra, Portugal.

2018 — **Organiser, Workshop: Logic for Children**, UNILOG 2018, Vichy, France.

Talks

SELECTED TALKS (INVITED AND/OR REFEREED)

Leiden University, The Netherlands, 2024	**109th Peripatetic Seminar on Sheaves and Logic**, The semantic lax descent factorization of a functor.
University of Santiago de Compostela, Spain, 2024	**International Conference in Category Theory 2024**, Doubly-infinitary distributive categories
University of Coimbra, Portugal, 2024	**SUMTOPO 2024, Topology and Logic in Computer Science**, Homomorphic reverse differentiation for partial features
University of Coimbra, Portugal, 2024	**38th Summer Conference on Topology and its Applications**, Free distributive and extensive categories
University of Coimbra, Portugal, 2023	**XIV Portuguese Category Seminar**, Freely generated bicartesian categories and exponentials
Université catholique de Louvain, Belgium, 2023	**International Conference in Category Theory 2023**, Spanning the tale of “Monades et descente”
Lorentz Center, Leiden, Netherlands, 2022	**New Challenges in Programming Language Semantics**, Categorical Approach to Semantic Logical Relations (for partial features and recursive types).
University of Coimbra, Portugal, 2022	**XIII Portuguese Category Seminar**, CHAD: elementary categorical aspects of automatic differentiation
University of Primorska, Slovenia, 2021	**8th European Congress of Mathematics**, Eilenberg-Moore, Kleisli and descent factorizations.
University of Coimbra, Portugal, 2019	**XII Portuguese Category Seminar**, Lax comma 2-categories.
University of Edinburgh, Scotland, 2019	**International Conference on Category Theory 2019**, Descent and Monadicity.
Stellenbosch University, South Africa, 2019	**Categorical Structures in Algebra, Topology and Logic**, Eilenberg-Moore, Kleisli and descent factorization.
University of Azores, 2018	**International Conference on Category Theory (plenary)**, Aspects of Descent via Bilimits.
Masaryk University, Brno, Czech Republic, 2018	**103rd Peripatetic Seminar on Sheaves and Logic**, Biadjoint Triangles and Applications.
Universidade de Santiago de Compostela, Spain, 2018	**102nd Peripatetic Seminar on Sheaves and Logic**, Non-canonical isomorphisms.
Université catholique de Louvain, Belgium, 2017	**5th Workshop on Categorical Methods in Non-Abelian Algebra**, Biadjoint Triangles.
University of Coimbra, Portugal, 2017	**XI Portuguese Category Seminar**, Freely generated n-categories and coinserters.
University of Coimbra, Portugal, 2016	**4th Workshop on Categorical Methods in Non-Abelian Algebra**, Bilimits Commutativity and Descent
University of Aveiro, Portugal, 2015	**International Conference on Category Theory 2019**, Kan extensions and descent theory.

(Invited 1-Hour Seminar) TALKS

IMPA, Brazil, 2025	**Algebra Seminar**, Biadjoint Triangles and Coherence
University of Cape Town, South Africa, 2025	**Category Theory Seminar**, Bifibrations, Spans, and the Bénabou-Roubaud Theorem
Stellenbosch University, South Africa, 2025	**Category Theory Seminar**, Two-dimensional monad theory, pseudodistributive laws, and doubly-infinitary distributive and extensive categories as pseudoalgebras
University of Cape Town, South Africa, 2025	**Category Theory Seminar**, Two-dimensional monad theory, and free extensive completions
University of Coimbra, Portugal, 2024	**Algebra, Logic and Topology Seminar**, Unraveling the iterative CHAD.
University of Coimbra, Portugal, 2022	**Algebra, Logic and Topology Seminar**, Descent for Split Fibrations.
Utrecht University, Netherlands, 2021	**Software Technology Group Seminar**, Automatic differentiation of expressive total functional languages.
University of Coimbra, Portugal, 2019	**Seminar of the Category Theory Group at the University of Coimbra**, On monadicity and descent.
University of Coimbra, Portugal, 2019	**Algebra, Logic and Topology Seminar**, Functorial Semantics and Descent II.
University of Coimbra, Portugal, 2018	**Algebra, Logic and Topology Seminar**, Functorial Semantics and Descent.
University of Coimbra, Portugal, 2018	**Algebra, Logic and Topology Seminar**, Biadjoint Triangles and Applications.
University of Coimbra, Portugal, 2018	**Algebra, Logic and Topology Seminar**, Towards semi left exactness and Janelidze-Galois within the lax idempotent context.
University of Coimbra, Portugal, 2016	**Algebra, Logic and Topology Seminar**, The deficiency of categories.
University of Coimbra, Portugal, 2015	**Algebra, Logic and Topology Seminar**, Biadjoint triangles, descent and coherence.
Universidade do Porto, Portugal, 2016	**Research Seminar Program**, Kan construction of adjunctions.
University of Coimbra, Portugal, 2016	**Research Seminar Program**, Homotopy Excision.
University of Coimbra, Portugal, 2013	**Joint PhD Program UC-UP Seminar**, Every prespectrum represents a homology theory.
Universidade de Brasília, Brazil, 2010	**Math Seminar**, Topological Dynamics and Applications in Number Theory (in Portuguese).
Universidade de Brasília, Brazil, 2010	**Math Seminar**, Dynamical Proof of the Van der Waerden Theorem (in Portuguese).

WORKSHOP TALKS

LAFI, London, UK	**LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024)**, Homomorphic Reverse Differentiation of Iteration.

Additional Information

EARLY-CAREER FELLOWSHIPS AND SCHOLARSHIPS

2020 – 2025	**PhD Fellowship (Promovendus)**, Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2013 – 2017	**PhD Fellowship**, National Council for Scientific and Technological Development (CNPq), Brazil.
2012 – 2013	**Doctoral Research Grant**, Centre for Mathematics, University of Coimbra, Portugal.
2009 – 2012	**Undergraduate and Master's Scholarships**, National Council for Scientific and Technological Development (CNPq) and IMPA, Brazil.
2005	**Selective Pre-University Academic Training**, EPCAR — Brazilian Air Force Preparatory School for Aeronautics, Brazil.

NATURAL LANGUAGES

Portuguese and English (willing to learn Danish and Swedish)